

Exact Finite Rigidity for Furter's $R(3)$: All Windows Through $n = 299$

Anonymous

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Abstract

Let $g_{k,3} \in \mathbb{Z}[x_1, x_2, x_3]$ be the coefficient of X^{k+1} in the compositional inverse of $X(1 + x_1X + x_2X^2 + x_3X^3)$. We prove, by exact computer-assisted certificates, that

$$\sqrt{(g_{d,3}, g_{d+1,3}, g_{d+2,3})\mathbb{Q}[x_1, x_2, x_3]} = (x_1, x_2, x_3) \quad (2 \leq d \leq 300).$$

Equivalently, Furter's polynomial-composition rigidity assertion $R(3, n)$ holds for every $1 \leq n \leq 299$. This extends the previously published finite computation for the relevant three-variable window from $d < 50$ to $d \leq 300$. Each new instance is certified at one prime by exact unit-ideal computations on all three strata of a weighted-projective cover; properness then transfers emptiness of the special fibre to characteristic zero. We also prove a periodic Kummer-theoretic zero-band theorem showing that no single prime can certify every degree by this route. Exact rational Euler–socle certificates through $d = 12$ and normalized-resultant data through $d = 25$ isolate two structural continuations, but universal $R(3)$ remains open. A frozen package supplies ordinary and optimized replay, independent coefficient reconstruction, negative controls, hashes, and a fresh-extraction verification receipt.

Status. This is an unrefereed computer-assisted theorem candidate. The exact finite claim is stronger than a probabilistic calculation, but the release has not received specialist peer review or an independent external reconstruction.

1 Introduction

Furter's rigidity conjectures ask whether an unexpectedly long initial cancellation in a polynomial composition forces both polynomials to be the identity. They are important because Furter connected them to closure relations between strata of plane polynomial automorphisms [1]. Edo and van den Essen found a further bridge to a restricted family in the Strong Factorial Conjecture [3]. The universal statement $R(3)$ is still open.

Lewis, Perry and Straub (LPS) recast related closure questions as explicit ideal computations in inverse-series coefficients and verified their stronger condition in the three-variable column for $d < 50$ [2]. The present work takes the radical-maximality half of that architecture and extends its exact finite rigidity consequence to $d \leq 300$.

Our principal result is deliberately finite.

Theorem 1.1 (Exact finite $R(3)$). *Furter's $R(3, n)$ holds for every integer n with $1 \leq n \leq 299$.*

The proof does not infer a characteristic-zero theorem from the apparent frequency of good modular reductions. For every degree it records an exact empty special fibre over a single finite field. A properness argument then gives the corresponding characteristic-zero conclusion. A bad prime has no negative implication for characteristic zero; it merely cannot certify that degree by this route.

The computation reveals a structural limitation. For every prime $p \geq 7$, infinitely many inverse coefficients vanish identically modulo p , so infinitely many three-term windows are necessarily bad at that prime. The exact map supplies a bad window for each of $p = 2, 3, 5$ as well. Thus no fixed prime certifies every degree, and any universal proof based on good special fibres must coordinate multiple primes.

2 Inverse coefficients and the rigidity dictionary

Give x_i weight i . Define $g_{n,3}$ by

$$g_{n,3} = \sum_{a_1+2a_2+3a_3=n} (-1)^{a_1+a_2+a_3} \frac{(n+a_1+a_2+a_3)!}{(n+1)n!a_1!a_2!a_3!} x_1^{a_1} x_2^{a_2} x_3^{a_3}. \quad (1)$$

These are integral weighted-homogeneous polynomials of weight n . Equivalently, if G is the unique formal solution with initial term w of

$$G = w - x_1 G^2 - x_2 G^3 - x_3 G^4, \quad (2)$$

then $g_{n,3}$ is the coefficient of w^{n+1} in G .

For $d \geq 2$ put

$$I_d = (g_{d,3}, g_{d+1,3}, g_{d+2,3}) \subset \mathbb{Z}[x_1, x_2, x_3].$$

Lemma 2.1 (Source dictionary). *The equality*

$$\sqrt{I_d \mathbb{Q}[x_1, x_2, x_3]} = (x_1, x_2, x_3)$$

is Furter's one-map assertion $\tilde{R}(3, d-1)$ and is equivalent to his two-map assertion $R(3, d-1)$.

Proof. The coefficient convention follows from Lagrange inversion and agrees with the LPS definition. Furter's Lemma 2 identifies the vanishing of the three consecutive inverse coefficients with $\tilde{R}(3, d-1)$ and proves the equivalence with $R(3, d-1)$ [1, 2]. The only common affine zero being the origin is precisely the displayed radical equality. $\square$

The shift $n = d - 1$ is essential: $2 \leq d \leq 300$ corresponds exactly to $1 \leq n \leq 299$.

3 From one good special fibre to characteristic zero

Let p be prime and write $I_{d,p}$ for the reduction of I_d in $\mathbb{F}_p[x_1, x_2, x_3]$.

Definition 3.1. The pair (d, p) is *GOOD* if $I_{d,p}$ has height three, equivalently if the quotient is finite-dimensional. It is *BAD* if the three forms have a nonzero common point over $\overline{\mathbb{F}}_p$.

Because the forms have weighted degrees $d, d+1, d+2$, a GOOD quotient is a weighted complete intersection. Its length is

$$\frac{d(d+1)(d+2)}{1 \cdot 2 \cdot 3} = \binom{d+2}{3} \quad (3)$$

and its socle weight is

$$d + (d+1) + (d+2) - (1+2+3) = 3d - 3. \quad (4)$$

Lemma 3.2 (Good-special-fibre bridge). *If (d, p) is GOOD, then $\sqrt{I_d \mathbb{Q}[x_1, x_2, x_3]} = (x_1, x_2, x_3)$.*

Proof. Regard the common-zero scheme as a closed subscheme of the weighted projective plane over $\text{Spec } \mathbb{Z}_{(p)}$. This morphism is proper. If its generic fibre were nonempty, the closure of a generic point would have closed image containing the generic point of $\text{Spec } \mathbb{Z}_{(p)}$, hence the closed point as well. The special fibre would then be nonempty, contrary to GOOD. Thus the generic weighted-projective fibre is empty, so the only affine common zero in characteristic zero is the origin. $\square$

This argument uses one prime for the whole projective fibre. Certificates from different primes on different affine charts cannot be combined.

For completeness, the same bridge has a graded-module formulation. With $A = \mathbb{Z}[x_1, x_2, x_3]/I_d$, every weighted piece A_k is a finite $\mathbb{Z}$ -module. A GOOD special fibre is a complete intersection with Hilbert series

$$\frac{(1-t^d)(1-t^{d+1})(1-t^{d+2})}{(1-t)(1-t^2)(1-t^3)}.$$

The special fibre vanishes above weight $3d - 3$. Nakayama's lemma then kills the corresponding localized integral pieces and makes the generic fibre Artinian. Equal complete-intersection Hilbert functions also show that the localized graded pieces are free, a fact used by the structural pilot to lift modular nonvanishing.

4 The exact finite computation

Two proof-bearing lanes cover the degree interval.

4.1 Complete modular map for $2 \leq d \leq 49$

For every one of the 48 degrees and every one of the 168 primes $p \leq 997$, an exact Singular Gröbner-basis computation classified the quotient as GOOD or BAD. The map contains 8,064 rows: 6,931 GOOD and 1,133 BAD. An independent formal-reversion implementation checks the generator formula on a fixed grid. The verifier rejects an impossible finite length and includes a mutation in which the third generator is replaced by zero.

This lane reproduces the radical half of the published LPS range. Its role in the present release is regression and continuity; the new finite range begins at $d = 50$.

4.2 Weighted-projective cover for $50 \leq d \leq 300$

Materializing complete quotient bases becomes increasingly expensive. The second lane instead certifies the absence of a nonzero projective common zero on the three strata

$$x_1 = 1, \quad x_1 = 0, \quad x_2 = 1, \quad x_1 = x_2 = 0, \quad x_3 = 1. \quad (5)$$

These strata cover the weighted projective plane over an algebraically closed field, including in characteristics 2 and 3. For a fixed attempt (d, p) , the producer reduces the integral generators at p and proves that the ideal on each stratum is the unit ideal. All three results are nested under the same prime-labelled attempt. A second specialize-first implementation rebuilds every selected GOOD certificate under ordinary and optimized Python.

The screen covers 251 degrees and records 2,854 attempts: 2,539 exact BAD fibres, 251 selected GOOD fibres, and 64 timeouts. A timeout has no mathematical meaning and is never used as evidence. The largest selected prime is 127, at $d = 280$; because some earlier attempts timed out, the selected prime is not always claimed to be the smallest possible GOOD prime.

Proposition 4.1 (Certified finite range). *For every d with $2 \leq d \leq 300$, there exists a recorded prime p for which (d, p) is GOOD.*

Computer-assisted proof. For $d \leq 49$, the exact modular-map receipt contains at least one row labelled GOOD for every degree and the verifier recomputes the expected length (3). For $50 \leq d \leq 300$, the assembled projective-screen receipt contains exactly one selected same-prime three-stratum unit-ideal certificate for every degree. Independent ordinary and optimized verifier receipts agree byte-for-byte. A root verifier binds the degree inventory, source and runtime hashes, controls, and subordinate receipts. A package verifier then extracts the frozen archive into a fresh directory, runs the replay, checks the manifest, and reproduces a PASS root. All computations are exact; no floating-point tolerance enters an acceptance path. $\square$

Proof of Theorem 1.1. Apply Proposition 4.1 and Lemma 3.2 for every $2 \leq d \leq 300$, then use Lemma 2.1 with $n = d - 1$. $\square$

Corollary 4.2 (Length-two Polydegree consequence). *For every $2 \leq k \leq 300$, Furter's closure equality of Theorem B holds for the length-two multidegrees $(4, k)$ and $(k, 4)$.*

Proof. Use $R(3, k - 1)$ in Furter's Theorem B and his symmetry lemma [1]. $\square$

This corollary is not presented as the first proof of the full $e = 3$ Polydegree column. A separate Evidence Press candidate has already proved that column for all degrees by a smooth-point/Fourier/interval-arithmetic route [7].

5 A fixed-prime obstruction

The modular data suggests blocks, but the key negative fact is analytic.

Theorem 5.1 (Periodic zero band). *Let $p \geq 7$ be prime and put $K_p = \lfloor (p + 2)/4 \rfloor$. If*

$$n \equiv -k \pmod{p}, \quad 2 \leq k \leq K_p,$$

then $g_{n,3}$ is the zero polynomial in $\mathbb{F}_p[x_1, x_2, x_3]$.

Proof. Consider a monomial indexed by (a_1, a_2, a_3) in (1), write $s = a_1 + a_2 + a_3$, and let b_i be the units digit of a_i in base p . The weighted-degree condition gives

$$b_1 + 2b_2 + 3b_3 \equiv -k \pmod{p}.$$

If $b_1 + b_2 + b_3 < k$, then the left side is at most $3(k-1) < p-k$, a contradiction. Thus $b_1 + b_2 + b_3 \geq k$. Adding the base- p units digits of n, a_1, a_2, a_3 therefore produces a carry, because the units digit of n is $p-k$. By Kummer's theorem the multinomial coefficient

$$\frac{(n+s)!}{n!a_1!a_2!a_3!}$$

is divisible by p . Since $n+1 \equiv 1-k \not\equiv 0 \pmod{p}$, division by $n+1$ preserves divisibility. Every coefficient of $g_{n,3}$ vanishes. $\square$

Corollary 5.2 (No universal fixed GOOD prime). *For every prime $p \geq 7$, infinitely many windows $I_{d,p}$ are BAD. No prime, including 2, 3, 5, supplies a GOOD witness for every degree through the good-special-fibre architecture.*

Proof. For $p \geq 7$, any three-term window intersecting the periodic zero band has at most two nonzero generators and thus height at most two. Explicitly,

$$p - K_p - 2 \leq d \pmod{p} \leq p - 2$$

forces BAD, and each congruence class recurs infinitely often. For $p = 2, 3, 5$, the exact map supplies a BAD witness at $d = 2, 3, 6$ respectively. Those three witnesses are sufficient for the stated no-universal-fixed-prime conclusion; no infinitude assertion for the small primes is needed here. $\square$

The same carry proof works for weights $1, \dots, e$. If $K_{p,e} = \lfloor (p+e-1)/(e+1) \rfloor$, then $g_{n,e} = 0 \pmod{p}$ whenever $n \equiv -k \pmod{p}$ and $2 \leq k \leq K_{p,e}$. We make no priority claim for this elementary generalization.

6 Structural evidence beyond the finite theorem

6.1 The canonical Euler–socle route

Define the weighted Euler matrix

$$(M_d)_{ij} = \frac{j}{d+i-1} \partial_{x_j} g_{d+i-1,3}.$$

Weighted Euler identities give

$$M_d(x_1, x_2, x_3)^T = (g_{d,3}, g_{d+1,3}, g_{d+2,3})^T.$$

Consequently the adjugate identity places $x_i \det(M_d)$ in I_d for each i , while $\det(M_d)$ has the socle weight $3d-3$. In an Artinian complete intersection the Jacobian represents a nonzero socle class [4].

This contracts one possible all-degree proof to two canonical identities. It would suffice to prove, for every $d \geq 2$, nonzero scalars λ_d, ν_d such that

$$x_1^{3d-3} - \lambda_d \det(M_d) \in I_d, \quad x_3^{d-1} - \nu_d \det(M_d) \in I_d. \quad (6)$$

Indeed the adjugate identities would force $x_1 = x_3 = 0$ on the common-zero locus, and the nonzero pure- x_2 Fuss–Catalan coefficient in the even member of each three-term window would then force $x_2 = 0$.

Exact rational certificates verify the three pure-power memberships, the one-lower nonmemberships, the Hilbert functions, the Euler identities, and (6) for $2 \leq d \leq 12$. This is a finite structural pilot, not an interpolation argument.

6.2 The finite-prime automaton route

Equation (2) defines an algebraic multivariate series over every finite field. Multivariate Christol theory therefore gives a finite p -kernel for its coefficient array [5, 6]. What is missing is not coefficient automaticity, but a finite-state lift from coefficient sections to an ideal certificate, such as a bounded family of leading-monomial or Gröbner signatures. In view of Corollary 5.2, a successful construction must use a product automaton for several primes and show that no reachable state is BAD in every component.

6.3 Calibrated negative results

Normalized weighted resultants were independently reconstructed through $d = 25$: full integer determinants through $d = 17$ and modular determinants with rigorous CRT reconstruction for $18 \leq d \leq 25$. Across 4,032 prime-degree comparisons, a prime divides the normalized resultant exactly when the corresponding fibre is BAD. A withheld-data screen excludes every first-order rational-ratio recurrence $R_{d+1}Q(d) = R_dP(d)$ with $\deg P + \deg Q \leq 12$. Separate screens reject a small d -independent raw-cofactor transport and low-complexity recurrences for the Euler–socle scalars. These results close only the specified ansatz classes; they do not rule out higher-order creative telescoping or a finite automaton.

7 Evidence object and assurance boundary

The release archive is an executable evidence object rather than a table of claimed CAS outputs. Its root receipt binds:

- the 299-degree inventory and the two exact certificate lanes;
- producer, verifier, input, runtime, and receipt SHA-256 identities;
- ordinary/optimized Python parity where applicable;
- independent formal reversion and sparse arithmetic on bounded pilots;
- generator-axis Catalan and Fuss–Catalan regressions through $g_{302,3}$;
- fail-closed controls for impossible colengths, missing generators, altered cofactors, incomplete prime prefixes, and malformed inventories;
- an exact counterexample gate with no numerical-tolerance path; and
- a fresh-extraction replay and manifest verification.

The large $d = 50, \dots, 300$ lane independently replays a second Singular wrapper, but it does not yet export portable sparse Nullstellensatz cofactors for all 251 degrees. Singular is therefore inside that lane’s trusted computing boundary. Internal cross-model adversarial review is not external specialist review, and executable replay is not formal verification.

8 Limitations and open problem

Theorem 1.1 is finite. It neither proves nor refutes the next window, and it supplies no monotonicity principle in d . Universal $R(3)$ would require an all-degree theorem, not a longer table. The two best-defined targets are:

1. prove both canonical identities (6) for every $d \geq 2$, including nonvanishing of their scalars; or
2. construct a finite set of primes and a certified product automaton that gives a same-prime GOOD fibre in every degree despite each prime's zero bands.

The companion Universal $R(3)$ Challenge records these obligations, exact finite examples, falsifiers, and acceptance criteria in machine-readable form.

Data and code availability

All proof scripts, receipts, controls, manifests, the frozen replay archive, and the machine-readable challenge are included with the public release. The archive is identified by SHA-256 in its manifest and external verification receipt. Public repository, release, and DOI links are inserted in the versioned record at publication.

Ethics statement

This work uses no human participants, personal data, animals, clinical data, or field interventions. No ethics approval was required.

CRediT contribution statement

Anonymous: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing, visualization, and project administration. The provenance record separately describes human research direction and model-assisted implementation and review.

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